Unit 19: Monotonic Functions and the First Derivative Test

1. The idea, in plain words

You already know that $f'(x)$ tells you the slope of f at each point. This unit turns that fact into a full map of a function's shape: where the slope is positive, the graph is climbing (increasing); where the slope is negative, the graph is falling (decreasing). That's really all "monotonic" means — climbing, or falling, over a stretch of the graph.

Picture it like a hiking trail again: if your elevation is always rising as you walk forward, the derivative (your steepness reading) stays positive the whole way. The moment you crest a hill and start heading downhill, your steepness reading flips to negative. The exact spot where you crest the hill — the transition point — is a critical point, and it corresponds to a local maximum. Cresting a valley (bottoming out and then climbing again) is a local minimum.

The First Derivative Test formalizes exactly this "cresting" idea: at each critical point, check whether $f'(x)$ switches from positive to negative (a local max — the graph was climbing, now it's falling), switches from negative to positive (a local min — the graph was falling, now it's climbing), or doesn't change sign at all (neither — just a flat "shoulder" where the graph pauses but keeps going the same direction).

The overall game plan:

1. Find $f'(x)$.
2. Find all critical points (where $f'(x) = 0$ or is undefined).
3. Use the critical points to chop the number line into separate intervals.
4. Pick one test point inside each interval and check the sign of f' there.
5. Positive sign = increasing on that interval; negative sign = decreasing.
6. At each critical point, compare the signs on either side to classify it as a local max, local min, or neither.

2. Toolbox

Increasing/Decreasing Test:

$$f'(x) > 0 \text{ on an interval} \implies f \text{ is increasing there}$$

$$f'(x) < 0 \text{ on an interval} \implies f \text{ is decreasing there}$$

First Derivative Test (for classifying a critical point c):

- If f' changes from positive to negative at c: local maximum at c.
- If f' changes from negative to positive at c: local minimum at c.
- If f' doesn't change sign at c (same sign on both sides): neither a local max nor min.

Building a sign chart (the practical tool): draw a number line, mark every critical point on it, then pick one convenient test value inside each resulting interval and plug it into a factored form of $f'(x)$ — factoring makes it easy to read off the sign of each piece without a full computation.

3. Common mistakes

- Testing only some of the intervals, or forgetting an interval entirely. Every critical point splits the line into one more region — make sure you test all of them.

- Sign errors when evaluating a factored expression at a test point. Take it slow: figure out the sign of each individual factor first, then multiply the signs together.
- Confusing "f is increasing" with "f' is increasing." These are two completely different statements — this unit is only about the sign of f', not whether f' itself is growing.
- Forgetting critical points where f' is undefined, not just where $f' = 0$ — a cusp or vertical tangent can still be a local max or min.
- Mixing up the sign-change directions. Positive-to-negative is a max (the graph was going up, now down — a peak); negative-to-positive is a min (a valley). It's easy to flip these under pressure — always double check against the "cresting a hill" picture.

4. Problem Set

For each problem, find the intervals where f is increasing and decreasing, and classify each critical point as a local max, local min, or neither.

Warm-up

1. $f(x) = x^2 - 4x + 1$
2. $f(x) = -x^2 + 6x - 5$
3. $f(x) = x^3 - 3x$
4. $f(x) = x^3 - 12x$
5. $f(x) = x^2 + 2x - 3$
6. $f(x) = 2x^3 - 6x$

Standard

7. $f(x) = x^3 - 3x^2 - 9x + 5$
8. $f(x) = x^4 - 4x^3$
9. $f(x) = 3x^4 - 4x^3 - 12x^2 + 1$
10. $f(x) = \dfrac{x}{x^2 + 1}$
11. $f(x) = (x-1)^2(x+2)$
12. $f(x) = x^5 - 5x$

Challenge

13. $f(x) = x^3 - \dfrac{3}{2}x^2 - 6x + 3$
14. $f(x) = x^{2/3}(x-5)$
15. $f(x) = \sin x + \cos x$ on $[0, 2\pi]$
16. $f(x) = x^4 - 8x^2 + 3$
17. $f(x) = \dfrac{x^2 - 4}{x}$

Applied

18. A company's profit (in thousands of dollars) is $P(x) = -x^3 + 15x^2 - 48x + 10$ for $x \geq 0$ units produced. Find where profit is increasing and decreasing, and identify the production level giving a local maximum profit.

19. A population (in thousands) is modeled by $P(t) = -t^3 + 9t^2 + 120$ for t in $[0, 10]$ years. Find where the population is increasing and decreasing, and identify when it reaches a local maximum.
20. A ball's height (in feet) is modeled by $h(t) = -16t^2 + 64t + 5$. Find where the height is increasing and decreasing, and identify the time of the local maximum height.
21. A greenhouse's temperature (in °F) is modeled by $T(t) = t^3 - 6t^2 + 9t + 20$ for t in $[0, 5]$ hours. Find where the temperature is increasing and decreasing, and identify the local max and local min temperatures.